

A. Shift Sort

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given a binary string* s of length n and you are allowed to perform the following operation any number of times (including zero):

- Choose 3 indices $1 \leq i < j < k \leq n$ and right shift or left shift the values on s_i, s_j, s_k cyclically.

For the binary string 110110, if we choose $i = 1, j = 2, k = 3$ and perform a right shift cyclically, the string becomes 011110; if we choose $i = 4, j = 5, k = 6$ and perform a left shift cyclically, the string becomes 110101.

Determine the minimum number of operations required to sort the given binary string.

* A binary string is a string that consists only of the characters 0 and 1.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 100$). The description of the test cases follows.

The first line of each test case contains a single integer n ($3 \leq n \leq 100$) — the length of the string.

The second line contains a binary string s of length n .

Output

For each test case, output a single integer — the minimum number of operations required to sort the given binary string.

Example

input	Copy
4 3 001 4 0110 6 110100 6 101011	
output	Copy
0 1 2 1	

Note

For the first test case, the given string is already sorted. So, no operations are needed.

For the second test case, we can choose $i = 1, j = 2, k = 4$ and perform a right shift cyclically. The string becomes equal to 0011, which is sorted.

Codeforces Round 1049 (Div. 2)

Contest is running

01:40:16

Contestant



→ Submit?

Language:

Python 3.13.2

Almost always, if you send a solution on PyPy, it works much faster

Choose file:

Choose File

 No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	464
Problem B	928
Problem C	1160
Problem D	1624
Problem E1	1624
Problem E2	928
Problem E	2552
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:18 from the first attempt

